

KKBOX:
Leading
Taiwan and
Asia-Pacific
digital music
platform
delivers
billions of
music streams
yearly

Tell us your challenge. We're here to help.

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ABOUT KKBOX



About KKBOX

KKBOX, Asia's leading music streaming service, was established in 2005 by a group of Taiwanese software programmers with a shared passion for technology and music. KKBOX features 45 million legal tracks and is currently available in Taiwan, Hong Kong, Japan, Singapore, and Malaysia with over 10 million users.

Industries: Media & Entertainment,
Gaming

Using Google Cloud Platform in Taiwan improved customer experiences for viewed music even complied with ob

to pay copyright for
music streams, and
supported expansion
into new markets.

Location: Taiwan

Google Cloud Results

- Reduces cloud infrastructure costs by 30% and system administration time by 10%
- Cuts system latency in half and achieves near 100% uptime, improving customer access to rapid, reliable services
- Supports a pan-Asian operation and new growth opportunities

Reduced
streaming
latency
milliseconds to
30 ms

Cloud CDN (<https://cloud.google.com/cdn/>),

BigQuery (<https://cloud.google.com/bigquery/>)
(<https://cloud.google.com/kubernetes-engine/>),

Compute Engine
(<https://cloud.google.com/compute/>)

Drive

Cloud Storage (<https://cloud.google.com/storage/>)

Cloud CDN (<https://cloud.google.com/cdn/>)

Kubernetes Engine
(<https://cloud.google.com/kubernetes-engine/>)

Google Workspace
(<https://workspace.google.com/>)

Google Drive
(<https://workspace.google.com/products/drive/>)

Gmail
(<https://workspace.google.com/products/gmail/>)

Calendar
(<https://workspace.google.com/products/calendar/>)

Founded by a group of music-loving software developers, KKBOX (<https://www.kkbox.com/>) has grown to become one of Asia's leading music entertainment companies. In 2005, the business launched one of the world's first music streaming services. Headquartered in Taipei, Taiwan, KKBOX has expanded to Hong Kong, Singapore, Malaysia, and Japan.

With a culture of reinvention and exploring new business models, KKBOX has expanded its business from music streaming to live events, technology services, content, and investments. The business aims to reinvent itself by pursuing innovation in digital entertainment.

The KKBOX streaming service features 45 million tracks and has more than 10 million members. KKBOX members can access music streams, exclusive videos, artist interviews, live concert reports, and entertainment news. They can also listen to the same tracks as their friends or their favorite artists, and chat with their friends at the same time.

"When KKBOX was founded, music streaming was new to the market – people were used to downloading MP3 files," says Hung-Yi Chen, Head of System Infrastructure, KKBOX. "However, they quickly embraced the concept of opening an application to listen to almost any track they want."

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*—Hung-Yi Chen, Head of
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In 2009, KKBOX expanded to Hong Kong and in December 2010, Japanese telecommunications business KDDI Group acquired most of the company's stock. Further investments followed from smartphone provider HTC in 2011 and from Singapore's sovereign wealth fund, GIC, in 2014. That same year, KKBOX launched KKTIX online ticketing and KKBOX Live online concert streaming.

Adding cloud services

KKBOX started operations with applications and services running in on-premises data centers. As the

business expanded across Asia, it began experiencing service reliability and performance challenges. “We had stored all our music in our data centers in Taiwan, and slow network connections meant listeners in Japan or Hong Kong experienced problems accessing our content,” explains Chen.

“As our customer numbers and streams grew, and we looked at new markets, we started looking for additional infrastructure to continue providing a high-quality service,” he adds. KKBOX initially turned to a traditional cloud service and content delivery networks to manage its capacity and performance issues.

However, continued growth and new events highlighted the need for KKBOX to again review its infrastructure. “We have an annual live music event, the KKBOX Music Awards, every year in January that we broadcast live on the internet,” says Chen. “In 2017 alone, this event recorded 11.7 billion unique clicks and attracted nearly 12 million fans from 30 media platforms watching live simultaneously. It’s a big challenge and we need a great deal of capacity to meet the demand.”

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Latency exceeded requirements

KKBOX used its incumbent cloud service to meet demand at the first staging of the event. However, with the cloud provider’s nearest regions located in Japan and Singapore, latency between Japan, Singapore, Hong Kong, and Taiwan exceeded business requirements at 50 to 60 milliseconds. “We believed if we could get down to 30 milliseconds for our streaming, it would be a far better experience for our consumers,” says Chen.

In 2014, KKBOX migrated its streaming service to the Google Cloud Platform (<https://cloud.google.com/>)

Region in Taiwan. The provider's technology team completed the move mostly using internal resources, with some assistance provided by Google. "I would ask Google the best or fastest way to complete a particular activity and they would give me the information we needed to do it ourselves," explains Chen.

BigQuery and Compute Engine key

KKBOX is using BigQuery.

(<https://cloud.google.com/bigquery/>) as a data warehouse to enable analysis of customer payloads – content consumed by customers – so it can determine the licensing fees that need to be paid, and to enable its business intelligence teams to query data so they can make decisions about products and strategies. Compute Engine

(<https://cloud.google.com/compute/>) infrastructure-as-a-service runs the KKBOX authentication, authorization, and single sign-on system that supports the company's internal applications.

KKBOX is now caching its content close to customers in different countries through Cloud CDN

(<https://cloud.google.com/cdn/>), accelerating performance and reducing serving costs. The business is also storing content in a more secure, reliable Cloud Storage

(<https://cloud.google.com/storage/>) environment.

“Google Cloud allows us to enhance the effectiveness of nearly 800 KKBOX users across six regions. Google Workspace in particular enables employees in different places to communicate and collaborate when they need to. Google Workspace is more stable and secure than an in-house system — and allows us to save 60 percent in personnel costs as well as 30

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and operating
costs.”

*—Eric Tsai, Director,
International, KKBOX and
President, KKStream*

Latency and costs cut

With key systems running in Google Cloud Platform in the Taiwan Region, KKBOX has reduced average latency to about 30 milliseconds while recording near-100 percent uptime for its streaming service – meeting the high expectations of customers. “Our costs with Google Cloud Platform are about 30 percent lower compared to our other cloud provider, which frees up budget for value-added activities,” says Chen. “In addition, we have reduced our system administration time by about 10 percent.”

Google Workspace powers collaboration

KKBOX is also using [Google Workspace](https://workspace.google.com/) (https://workspace.google.com/) to cut administration loads and to foster a collaborative culture. “We first introduced Google Workspace because we didn’t want to manage our own mail server and because we needed the shared storage available on [Drive](#)

(<https://workspace.google.com/products/drive/>)," explains Chen. "Using Gmail (<https://workspace.google.com/products/gmail/>) and Calendar (<https://workspace.google.com/products/calendar/>) to manage scheduling – personally and for our meeting rooms – and Drive to store and share files, has fostered a highly collaborative culture in our business."

"Google Cloud allows us to enhance the effectiveness of nearly 800 KKBOX users across six regions," says Eric Tsai, Director, International, KKBOX and President, KKStream. "Google Workspace in particular enables employees in different places to communicate and collaborate when they need to. Google Workspace is more stable and secure than an in-house system – and allows us to save 60 percent in personnel costs as well as 30 percent in infrastructure maintenance and operating costs."

KKBOX is now preparing to run some applications in Google Kubernetes Engine (<https://cloud.google.com/kubernetes-engine/>). "We've chosen Kubernetes Engine because it is a new service and as a business we're keen to try cutting-edge technologies," explains Chen. "We're currently working up some use cases for Kubernetes Engine to take advantage of the improved manageability and scaling it offers."

The business is also exploring the potential of Google artificial intelligence, machine learning, and image and video recognition services to build an

intuitive recommendation engine. “Google Cloud has opened up a range of opportunities for KKBOX,” says Chen. “We look forward to working closely with Google in future.”